

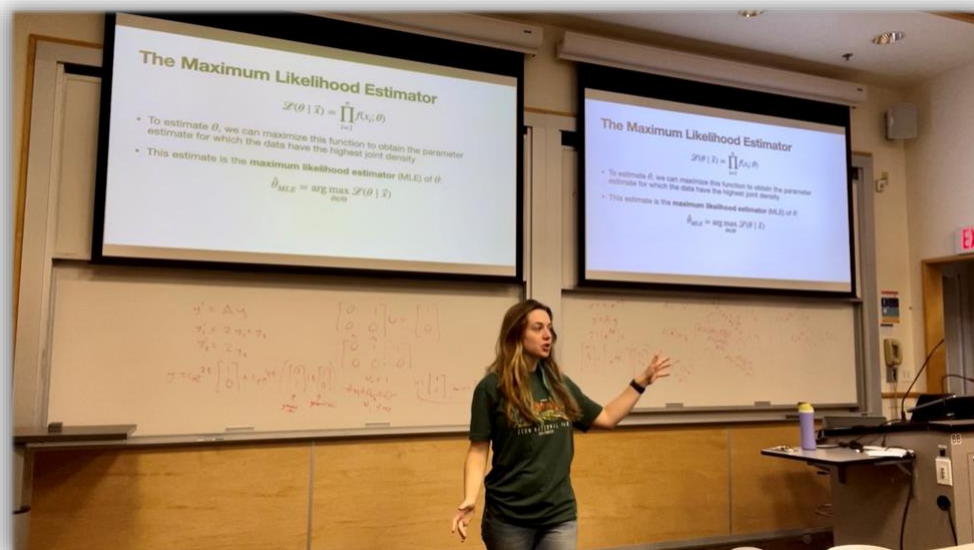
Teaching Portfolio

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Note: student comments and survey responses are reproduced anonymously, and quotations retain students' original wording except where minor edits are indicated.



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1 Teaching Philosophy & Introduction to the Document

As a child of two public school teachers, I have been thinking about teaching my entire life. At six, I would come home from kindergarten and talk with my dad about why my teacher had asked me to read aloud to the class; at fifteen, my mom and I would critique my biology teacher's assessment style on the drive home from school. Over time, I too became a teacher – starting with tutoring high school calculus, all the way up to teaching my own courses at UCI. Through the years, I have always returned to thinking about the same challenge: how do we hold students to rigorous standards of learning while still prioritizing compassion? I once heard a presenter at an education seminar joke that “students are a lot like people.” They are human; they are exhausted and stressed, but we are still responsible for training them to be rigorous statistical thinkers rather than doing them the disservice of making their classes easy.

My broader philosophy of teaching rests on the belief that compassion and rigor are not competing goals; that is, my role is not to remove the difficulty inherent in statistical reasoning, but to give students the support and sense of purpose they need to engage seriously with it and develop into confident and independent statistical thinkers. An important part of creating an inclusive and accessible classroom is distinguishing between the intellectual difficulty inherent to statistics and mathematics, and the barriers that arise from differences in students' prior preparation. I want all students to be able to engage with the former without being unnecessarily limited by the latter. This shapes my teaching in several ways: (1) I use rigorous assessment but pair it with clear expectations; (2) I situate problems in socially meaningful or scientifically relevant contexts that show students what mathematics and statistics are *for*; (3) I engage students in authentic applied thinking, including the choices and ambiguity that characterize math as used in practice; and (4) I try to create a nonjudgmental environment where all students feel able to work through uncertainty and take intellectual risks.

I want students to become active participants in statistical and mathematical reasoning rather than passive users of tools and procedures. What that requires changes across courses; in an introductory statistics course, students may be learning to reason critically about evidence and uncertainty for the first time, while in a more mathematically oriented course, they may use formal derivations and mathematical properties to understand why statistical methods work and what their results imply. Across these settings, however, my goal is the same: to ask students to do meaningful intellectual work while providing the clarity, support, and classroom environment they need to take on that challenge.

The remainder of this portfolio illustrates how I put these principles into practice across courses with very different statistical and mathematical demands and with students from widely varying mathematical backgrounds. In Section 2, I describe my teaching and mentoring experience, including both introductory courses and mathematically rigorous undergraduate and graduate coursework. In Section 3, I present representative course materials that demonstrate my practical approaches to the tenets I describe above, followed by evidence of teaching effectiveness from student feedback and reflection in Section 4. Finally, in Section 5, I describe how I continue to develop as an educator through student feedback, engagement with scholarship in statistics and mathematics education, and an interest in education research.

2 Teaching Experience and Responsibilities

This section gives a concise picture of the range, scale, and level of my teaching responsibilities. The full record remains in my curriculum vitae; here I highlight the experiences most relevant to my development as an instructor in the mathematical sciences. In the following subsections I describe the structure of each course in further detail.

Table 1: Selected formal teaching experiences.

Course / experience	Role & audience	Scale	Relevant responsibilities
STATS 8: Introduction to Biological Statistics	Instructor of record; undergraduate biological sciences students	211 enrolled	Course design (website, assessments, activities, lectures, etc.) and delivery; coordination of 3 TAs + 1 grader
STATS 67: Introduction to Probability and Statistics for Computer Science	Instructor of record; undergraduate computer science and engineering students	24 enrolled in Summer 2023; 13 enrolled in Summer 2026	Course design and delivery; assessments; applied computing examples; development of course materials
STATS 230: Statistical Computing	Teaching assistant; core graduate course for PhD and MS students in Statistics	15 enrolled in Winter 2026; 16 enrolled in Winter 2025	Student support; grading and feedback
STATS 211: Generalized Linear Models	Teaching assistant; core graduate course for PhD students in Statistics	20 enrolled	Student support; grading and feedback
Summer Institute in Biostatistics (SIBS)	Research mentor; undergraduate students in a 6-week intensive summer program	4 mentees	Mentored applied collaborative research

2.1 As Instructor of Record

2.1.1 For STATS 67

My first experience as instructor of record was with STATS 67, Introduction to Probability and Statistics for Computer Science, which I taught in Summer 2023 (and again in Summer 2026). This was an intensive five-week summer course covering sampling and study design, probability, univariate and multivariate random variables and distributions, statistical inference, and simple and multiple linear regression. I taught 24 students in 2023 and co-taught 13 students in 2026 with a peer. In 2023, as the sole instructor (without co-instructors, teaching assistants, or graders), I was responsible for all aspects of course design and delivery.

This course is intended for, but not restricted to, computer science students, and it combines mathematical development with applied statistical reasoning. Students are expected to do probability calculations and derivations requiring single- and multi-variable calculus and some linear algebra and connect these concepts to statistical inference and reasoning. For example, in one STATS 67 exam question, students were asked to

derive estimators, comment on their properties, and use those properties to make statistical judgments where there may be no single correct answer (Figure 7). Because the summer term compresses a full course into five weeks, I structured the course around frequent opportunities to practice and receive feedback. In 2023, students completed weekly homework and timed take-home assessments, with a weekly discussion section devoted primarily to computation and simulation. I ultimately moved away from take-home assessments (Section 5.1) and gave three in-class exams in 2026. Selected student feedback and materials from STATS 67 appear in Sections 3, 4, and 5.

2.1.2 For STATS 8

In Spring 2026, I taught STATS 8, Introduction to Biological Statistics, with 211 enrolled students. The course introduces students in the biological sciences to study design, data visualization and description, probability, statistical inference, and basic regression ideas. As instructor of record, I was responsible for the overall design and delivery of the course, including lectures, assessments, discussion activities, the course website, and computational materials. I also coordinated an instructional team of three teaching assistants and one grader, who delivered the discussion sections, held office hours, and graded exams and homework.

The scale and audience of STATS 8 shaped the course differently from STATS 67. Many students entered with no statistical or programming experience and the course needed to prepare them to interpret and conduct statistical analyses in later scientific coursework. It was important to me to maintain the statistical authenticity as much as possible in spite of this, for example by developing a point-and-click R code tool (Figure 9) and delivering mathematical mini-lectures to catch students up on the math skills they would need for the course. I also designed discussion activities around open-ended statistical questions rather than procedural exercises, for example, designing a study as an agent with bad intentions (Figure 8).

Assessment in STATS 8 combined explicit expectations with questions requiring students to apply concepts in unfamiliar contexts. Before each exam, students received a proficiency checklist specifying what they should be able to do; exams then asked them to apply those proficiencies to scientific studies, data displays, regression problems, and claims encountered in the media (Section 3.1).

Coordinating a course of this size also required me to think beyond my own classroom interactions to ensure that my TAs, grader, and I were all on the same page. We met once each week to talk about discussion sections, exams, and general logistics, and kept up active email communication.

2.2 As a Graduate-Level Teaching Assistant

My teaching experience also includes serving as a teaching assistant in core graduate courses in the UCI Statistics Ph.D. and M.S. programs. I was a teaching assistant for STATS 230, Statistical Computing, in Winter 2025 and Winter 2026, and for STATS 211, Generalized Linear Models, with classes of approximately 15-20 graduate students. My responsibilities included grading, providing detailed feedback on statistical reasoning and computation, and supporting students as they worked through advanced course material.

These courses have given me experience teaching students at a substantially different stage of statistical development from those in STATS 8 and STATS 67. In graduate courses, my feedback can assume greater mathematical and computational maturity and focus more heavily on the quality of students' statistical reasoning, interpretation, implementation, and rigorous mathematical understanding. This experience complements my introductory teaching and has prepared me to support students as they progress toward more advanced statistical study.

2.3 As a Research Mentor

My teaching also extends beyond formal coursework into undergraduate research mentoring. Through an NIH-funded Summer Institute in Biostatistics (SIBS) housed at UCI, I mentored four undergraduate students during an intensive six-week collaborative research project examining tuberculosis treatment outcomes in Botswana. I worked closely with the students on statistical analysis, coding, interpretation, literature review, and writing and presentation, while helping them navigate the less structured process of answering a scientific question for which neither the analysis nor the answer was known in advance. The project ultimately developed into a manuscript posted on medRxiv (Baniasad et al., 2025).

Research mentoring represents the far end of the progression toward statistical independence that I describe in my teaching philosophy. Rather than learning a predetermined set of methods, students must decide what questions to ask, determine which analyses are appropriate, interpret ambiguous results, and communicate their reasoning to collaborators. My role then shifts from explaining and structuring material toward asking questions, providing feedback, and helping students develop confidence in their own statistical judgment.

2.3.1 Broadening Participation in the Mathematical Sciences

My mentoring has also included programs designed to broaden participation in the mathematical sciences. During SIBS, I created a Slack channel without faculty present where students could work through tentative ideas and questions with me before bringing them to the larger research group. I have taken a similar approach in informal research mentoring and in my work as a peer mentor through UCI's Diverse Educational Community and Doctoral Experience (DECADE) program and the Pacific Alliance for Low-income Inclusion in Statistics and Data Science (PALiSaDS). Across these settings, I try to make asking for help and being uncertain feel like ordinary features of learning and research rather than evidence that a student does not belong.



Figure 1: SIBS students giving their final presentation at the end of Summer 2025.

3 Selected Teaching Practices and Evidence

Section 1 describes the four main principles that guide my teaching. The following section illustrates what those principles look like in practice through a small number of representative teaching materials, each accompanied by a brief explanation of its purpose and use.

3.1 Assessment with clear expectations

I use assessment to direct students toward the learning I value while making expectations as explicit as possible. I want assessments to be challenging because they ask students to demonstrate difficult skills, not because students are uncertain about what they are expected to know or be able to do. In STATS 8, I put this into practice through detailed Proficiencies Checklists that identify the skills students should be prepared to demonstrate on each exam.

Figure 2 pairs selected items from the Proficiencies Checklist for the first midterm with example exam questions that assessed them. I intentionally use action verbs, for example, “identify,” “interpret,” “distinguish,” and “explain”, so that expectations are expressed in terms of skills students should be able to demonstrate rather than topics they should memorize. Not every proficiency necessarily appears on an exam, but *every exam question corresponds to one or more stated proficiencies*. This way, students do not know exactly what questions they will encounter, but they do know what they will be expected to be able to do. This allows me to use closed-book, closed-note exams that require students to demonstrate their knowledge while reducing uncertainty about what constitutes adequate preparation.

STATS 8 Midterm 1 Proficiencies

A student who is prepared for the exam should be able to do each of the following, without relying on the wording of a particular example.

Sampling and bias

- ☐ Identify and distinguish simple random, stratified, cluster, systematic, convenience, and volunteer sampling from a description of how a sample was collected.
- ☐ Identify the sampling frame and observational units in a study.
- ☐ Recognize response bias, nonresponse bias, and selection bias from realistic examples.
- ☐ Explain why a sampling method or source of bias may affect how well a sample represents the population of interest.

Study design and variables

- ☐ Distinguish observational studies from experiments.
- ☐ Identify common observational study designs discussed in class given a scenario.
- ☐ Identify common experiments discussed in class given a scenario.
- ☐ Recognize single- and double-blinding and explain who is unaware of treatment assignment.
- ☐ Classify variables as nominal, ordinal, or quantitative.

Samples, populations, and graphical summaries

- ☐ Read and interpret a graph of categorical data, including identifying what its axes represent.
- ☐ Determine how many variables are represented in a graph or study description.
- ☐ Distinguish a sample from a population and a statistic from a parameter.
- ☐ Use information about how a sample was obtained when deciding what conclusions are supported.

Linear regression and association

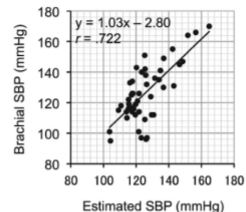
- ☐ Identify the explanatory and response variables in a regression setting.
- ☐ Understand where the least-squares line came from (you do not need to be able to calculate the coefficients).
- ☐ Read the slope and intercept of a least-squares regression line from regression output or a displayed equation.
- ☐ Interpret a regression intercept in context and recognize when the intercept does not have a useful practical interpretation.
- ☐ Interpret the slope of a regression line as an association between the explanatory and response variables.
- ☐ Interpret R-squared as the percentage of variability in the response variable explained by the linear relationship with the explanatory variable.
- ☐ Use a regression line to make a prediction for an appropriate value of the explanatory variable.
- ☐ Recognize extrapolation.

Drawing conclusions from studies

- ☐ Distinguish association from causation.
- ☐ Determine whether a study design supports a causal conclusion.
- ☐ Evaluate whether a media headline or scientific claim accurately represents the results of a study.

5. (12 points) Blood pressure (BP) cuffs are the clinical standard for accurately measuring blood pressure. A group of researchers in a 2018 study explored a “cuff-less” method of estimating BP using only a smartphone. They then looked at the association between estimated systolic BP (using the smartphone) and brachial systolic BP (using a cuff) for a group of study participants. A scatterplot of the estimated systolic BP and brachial systolic BP is given below.

Note that “brachial” just means “involving the arm”.



(a) What is the intercept of this regression line? _____

(b) Does the intercept of the regression line have a clear interpretation? If so, what is it; if not, why? _____

(c) What percentage of the variation in brachial SBP can be explained by the SBP estimated by the researchers' smartphone method? Round to one decimal place (e.g., 12.3%). _____

(d) Suppose the researchers collect data from one additional individual whose values for estimated SBP and brachial SBP fall exactly on the current least-squares regression line. How would including this new observation affect the regression line?

- ☐ The regression line would remain the same.
- ☐ The regression line would change slightly, but we cannot tell how from the information given.
- ☐ The slope of the line would increase.
- ☐ The slope of the line would decrease.

(e) For a person whose estimated SBP is 140, use the regression line to predict their brachial SBP. Round to one decimal place. Write NA if you think you cannot make a valid prediction with the given information. _____

(f) For a person whose estimated SBP is 80, use the regression line to predict their brachial SBP. Round to one decimal place. Write NA if you think you cannot make a valid prediction with the given information. _____

6. (4 points) A cohort study was published in early 2026 in which researchers recorded the participants' consumption of virgin olive oil and common olive oil, the diversity of their gut bacteria, and their performance on cognitive tests over the course of 2 years. Overall, participants who consumed more virgin olive oil tended to have greater diversity of gut bacteria and better cognitive test scores. A Fox News headline reporting on this study is shown below.

HEALTH

One type of olive oil has a surprising effect on brainpower during aging

Extra virgin olive oil found to improve cognitive function through increased gut bacteria diversity

By Khloe Quill - Fox News

Published February 25, 2026 6:10pm EST

The headline states, "One type of olive oil has a surprising effect on brainpower during aging". The subheadline says, "Extra virgin olive oil found to improve cognitive function through increased gut bacteria diversity".

Do you think the headline and subheadline appropriately represent the results of this study?

- ☐ Yes, because the study was observational and the headline does not imply a causal relationship between virgin olive oil and cognitive performance.
- ☐ Yes, because the study was experimental and supports the headline's claim of a causal relationship between virgin olive oil and cognitive performance.
- ☐ Yes, because though the study was observational, the causal mechanism proposed by the researchers is plausible so we can conclusively support the headline's causal claim.
- ☐ No, because it is implausible that virgin olive oil is associated with cognitive function, so the researchers' result is likely erroneous.
- ☐ No, because the study was observational and the headline implies a causal relationship between virgin olive oil and cognitive performance.

3. (8 points) Hyperpigmentation is a primarily cosmetic condition in which a person's skin tone is uneven, and there are various treatments available to lighten it. The biotechnology company AbbVie did a study to see if there was evidence that their new serum worked as well as the standard cream for treating hyperpigmentation.

1 (a) The study conducted by AbbVie used a "split-face" design, in which each participant was treated with the new serum on one half of their face and the standard cream on the other half. The sides of the face were randomly assigned, and the effects of both treatments on each person were compared after 12 weeks of treatment. Which study design best describes this approach?

- ☐ case-control observational study
- ☐ cross-sectional observational study
- ☐ cohort observational study
- ☐ completely randomized experiment
- ☐ matched pairs experiment

2 (b) Neither the study participants nor the researchers knew which side of their face was receiving which treatment until the study was over. What kind of blinding is this?

- ☐ double dummy
- ☐ double-blind
- ☐ single-blind
- ☐ none of the above

3 (c) At the end of treatment, a number of variables were collected from each individual in the study. Identify each variable below as ordinal, nominal, or quantitative.

_____ : clinical grading of hyperpigmentation as none, mild, moderate, or severe

_____ : irritation types as erythema, edema, dryness, burning, stinging, and itching

_____ : health-related quality of life scores, on a scale from 0 to 60

Figure 2: Sample STATS 8 exam questions and Proficiencies Checklist.

After the exam, over 90% of students rated the expectations either "clear" or "very clear" on a post-midterm survey (Figure 3). This clarity showed up in my end-of-term evaluations as well; in STATS 8 95% of students agreed or strongly agreed that I presented material in a way that helped them understand it. Importantly, this did not come at the cost of rigor; only 15% described the course as easy (see Section 4 for further evidence of teaching effectiveness). This is the response I hoped for; rigorous exams should challenge students to meet high expectations, not to decipher them.

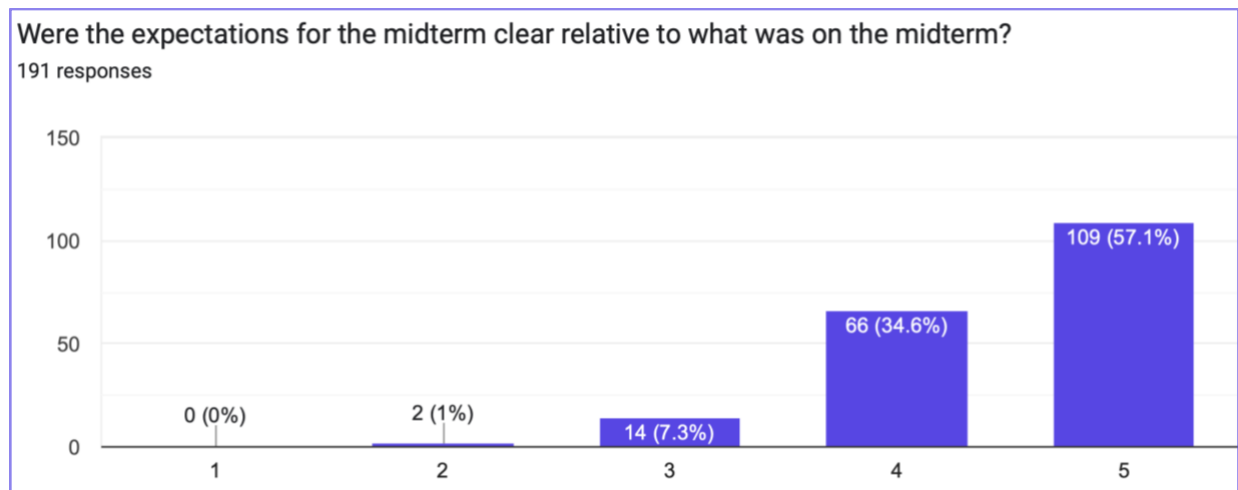


Figure 3: Results from post-exam survey.

3.2 Meaningful contexts

I also believe that putting technical methods into interesting contexts makes students more willing to grapple with difficult ideas, helping them to enjoy the course more without lowering intellectual demand. Context is not an embellishment added after students learn a statistical procedure; it helps establish what the procedure allows them to learn and why the underlying statistical reasoning matters. John Tukey put it well: "The best thing about being a statistician is that you get to play in everyone's backyard." I try to bring that spirit into class by tailoring examples to my students' interests and backgrounds, and I also develop reusable teaching materials that connect clearly and directly to consequential real-world questions.

3.2.1 Learning what matters to my students

Meaningful examples depend in part on knowing who my students are. Before the first class meeting in STATS 8 and STATS 67, I ask students to complete a short welcome survey about their backgrounds, interests, previous experiences with statistics, and concerns about the course. I use these responses both to understand the class I am teaching and to identify contexts that can make otherwise abstract statistical ideas more recognizable (Figure 5). For example, in STATS 8, 25 students identified sports as among the things they are most passionate about, which led me to incorporate examples involving baseball, basketball, sports-related injuries, etc. (for example, Figure 4). In STATS 8, a course designed primarily for biological sciences students, I also drew heavily on biomedical and scientific studies, including observational biological data, clinical trials results, and racial bias in healthcare. In STATS 67, I included engineering and programming examples.

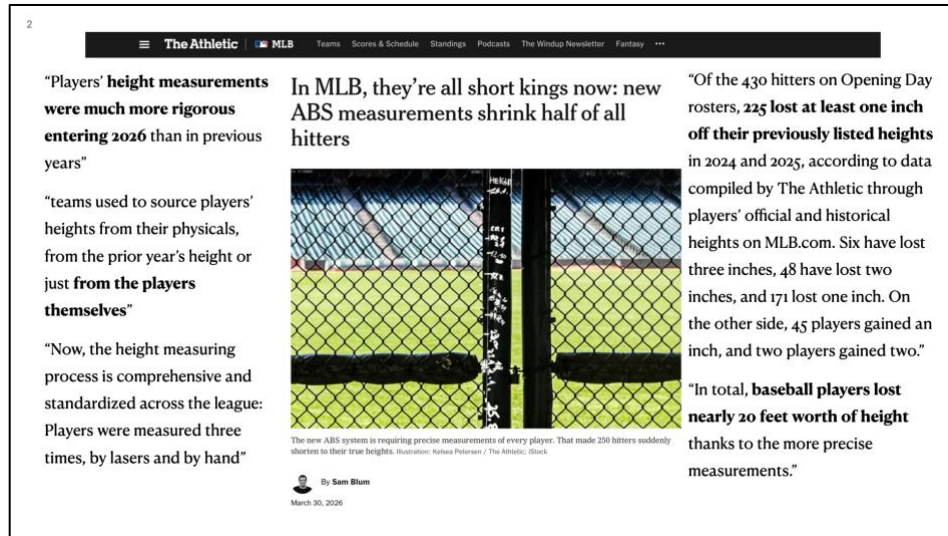


Figure 4: Lecture slide of response bias example in the context of recent baseball news.

STATS 8 Welcome Survey

Welcome to STATS 8! This survey asks questions that will help the teaching team support you in our course. Feel free to leave questions blank if you don't feel comfortable answering them. Your responses will only be seen by Jessalyn and the TAs for the course.

* Indicates required question

Email *

☐ Record jessalys@uci.edu as the email to be included with my response

Name *

Include information about the name you would like to be called or the pronunciation of your name if you would like to!

Your answer

What major are you in? *

Your answer

What are you most passionate about?

Your answer

Do you have any food allergies or intolerances? If so, please list them below.

Your answer

Which discussion section are you in? *

☐ 9am

☐ 10am

☐ 1pm

☐ 4pm

☐ Other:

How much prior knowledge (from school or other sources) do you have about statistics? *

All knowledge levels are totally okay! This is to help us calibrate our teaching to the needs of the class.

☐ I feel comfortable with basic algebra (solving simple equations, working with formulas, percentages, proportions, averages).

☐ I can understand graphs like histograms, scatterplots, or bar charts.

☐ I have taken a statistics course before.

☐ I have learned about probability (e.g., dice, coins, basic rules).

☐ I have learned about probability distributions (pmf, pdf, discrete, continuous, specific distributions like normal, binomial, etc.)

☐ I have learned about hypothesis tests or confidence intervals.

☐ I have learned about correlation and regression.

☐ I have used a programming language before (any language).

☐ Other:

Why are you taking this course? Select all that apply.

☐ Major requirement

☐ Minor requirement

☐ Gen ed requirement

☐ Interested in the subject

☐ Need credits

☐ Other:

If there are reasons you will occasionally need to miss class (sports participation, for example), please explain.

Your answer

How are you feeling as we enter this quarter?

☐ Great!

☐ Good

☐ Okay

☐ Not very good

☐ Very bad

Do you have access to the following resources? *

☐ A place to live

☐ Adequate food

☐ Necessary medications

☐ A good place to study

☐ A stable internet connection

☐ Emotional and social support

Is there anything happening in your life right now that makes it hard for you to focus on school? This can include being ill, caring for an ill loved one, experiencing a personal hardship, etc.

Your answer

Are there any specific forms of support that would help you succeed in the course? These could look like many things, ranging from exam review sessions to space in class to talk about mental health.

Your answer

What concerns do you have about the course at this point? Are there any aspects of the course you are excited about?

Your answer

Submit Clear form

Figure 5: Example welcome survey given to students at the beginning of the term.

3.2.2 Development of the *deitaset*s Package

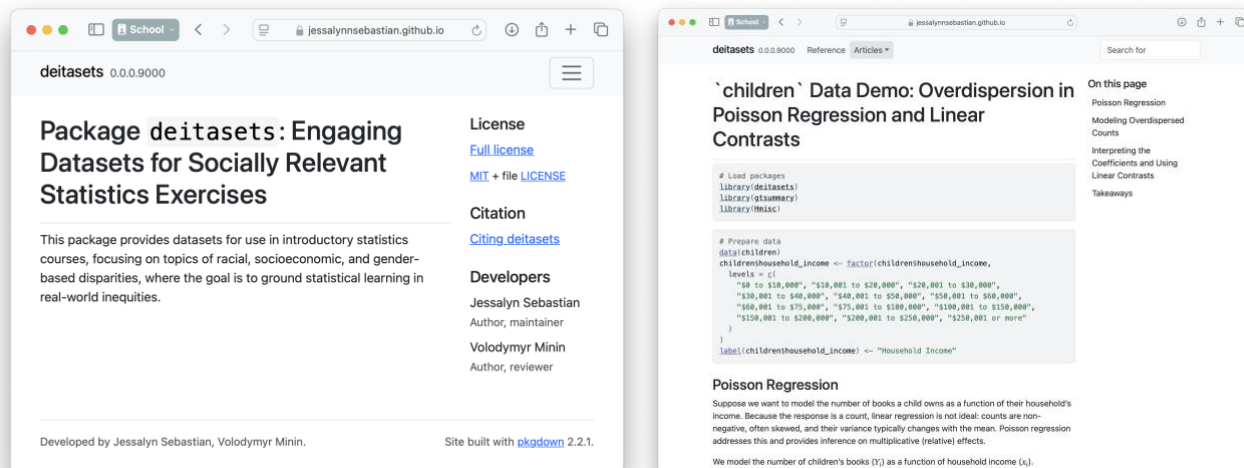


Figure 6: Screenshots from the website associated with the *deitaset*s package.

My interest in meaningful contexts has also extended beyond the design of individual courses. Through the UCI Division of Teaching Excellence and Innovation’s Summer Scholar program, I began developing *deitaset*s, an R package that collects datasets with contemporary social and political relevance for use in statistics instruction. The project grew out of the practical difficulty of finding datasets that are both pedagogically manageable and connected to questions students recognize as consequential. Rather than constructing artificial examples, I clean and curate data from publicly available studies and pair them with instructional materials demonstrating the statistical questions they can motivate. For example, one vignette uses children’s book ownership across household income levels to motivate count regression, overdispersion, model diagnostics, and linear contrasts. The accompanying vignettes are intended to make these materials readily adaptable for classroom use. The goal of *deitaset*s reflects the same principle that guides my own teaching; context should not be an afterthought added to an otherwise abstract exercise, but a way for students to understand what statistical reasoning allows them to learn about the world.

3.3 Authentic thinking

I also believe that teaching honestly – that is, the way that methods are used in practice, without sanitizing them or erasing ambiguity – makes students more willing to engage with challenging material. I try to give students opportunities to work through ambiguity and justify decisions, while calibrating the form of that work to their mathematical, statistical, and computational preparation.

In STATS 67, students have enough mathematical background to engage directly with the foundations of statistical methods. The assessment question shown in Figure 7 asks students to derive and compare two estimators of the endpoint of a Uniform distribution. After deriving their properties, students are shown their bias and variability across sample sizes and asked which estimator they would prefer. I explicitly tell students that there is not necessarily one correct choice, because the purpose is to use the mathematical results they have derived to make and justify a statistical judgment.

1. A plant biologist is studying the time required for seeds of a particular species to germinate under controlled laboratory conditions. As a simplified model, suppose the germination time X , measured in days, has probability density function

$$f(x | \theta) = \frac{1}{\theta}, \quad 0 \leq x \leq \theta, \quad \theta > 0.$$

Thus, θ represents the longest possible germination time under this model. Suppose $X_1, \dots, X_n$ are independent germination times from this population.

We will consider two different estimators of θ : one obtained using the Method of Moments and one obtained using maximum likelihood.

- (a) Use the Method of Moments (MoM) to find an estimator of θ .
- (b) Is $\hat{\theta}_{\text{MoM}}$ biased or unbiased? Justify your answer.
- (c) Now consider maximum likelihood estimation. Write down the likelihood function $L(\theta)$. You are told that

$$\hat{\theta}_{\text{MLE}} = \max(X_1, \dots, X_n).$$

Using the likelihood, explain conceptually why the maximum observed germination time maximizes the likelihood.

- (d) Let

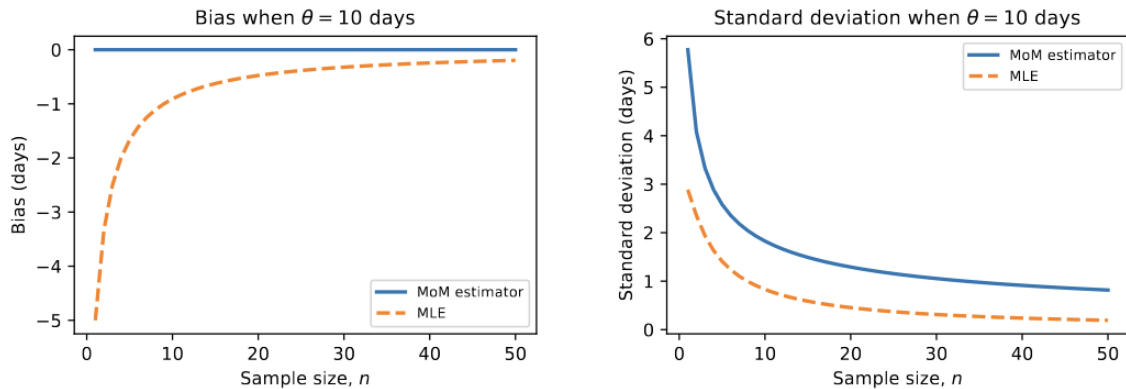
$$X_{(n)} = \max(X_1, \dots, X_n).$$

You are given that

$$E(X_{(n)}) = \frac{n}{n+1}\theta.$$

Find the bias of $\hat{\theta}_{\text{MLE}}$. Then, explain conceptually why the direction of this bias makes sense in this context.

- (e) Suppose the true maximum germination time is $\theta = 10$ days. The figures below show the bias and standard deviation of the two estimators for different sample sizes.



Under this hypothetical scenario where $\theta = 10$, discuss the advantages and disadvantages of using the MoM estimator versus the MLE. Your answer may depend on the sample size, and there is not necessarily one correct choice. Justify your reasoning.

Figure 7: Sample STATS 67 exam question.

Students in STATS 8 generally have much less mathematical preparation, but they can still begin practicing the judgment involved in statistical work. Discussion activities therefore often replace prescribed exercises with problems that leave some aspect of the analysis for students to decide. In the example activity shown in Figure 8, students work backward from a desired conclusion and design a study that would make that conclusion more likely. Rather than reproducing a procedure demonstrated in lecture, students must draw on their understanding of sampling, measurement, and bias to make and justify their own study design choices. There is no single

correct design. Students must identify different choices they could make about how data are collected and reason about how those choices might systematically influence the result. One student later reflected that these activities made them realize that “a strong foundation in stats is absolutely fundamental... because it’s a particular way of thinking.”

The deliberately mischievous premise also serves a serious purpose. Seeing how easily they can construct a persuasive but biased study themselves helps students understand why study design matters outside the classroom and why statistical claims cannot be evaluated independently of how the data were produced.

Discussion 2 Wednesday, April 8, 2026 Complete this worksheet in a group of 3-5 students. Write your answers by hand, then take a picture and submit it to Canvas. When you submit, only one person from your group needs to submit, but make sure to specify who is in the group. Select one of the scenarios below, and specify which scenario you are choosing. For your choice of scenario, answer the following. - Select one scenario (1, 2, or 3). - What is your sampling frame? Who are your sampling units? - How do you collect your sample, logistically? What type of sampling is it? - What information do you collect? How do you measure it? - Is your study observational, or experimental? Why? - Explain why the way you have chosen to design your study will support your desired conclusion. - How would you alter the study to answer the question of interest in a more honest way?	Scenarios (pick one of the three) Scenario 1: Sugar Industry This is adapted from documented Sugar Research Foundation efforts in the 1960s to emphasize fat and cholesterol as the main dietary causes of coronary heart disease while downplaying sugar. A historical analysis of internal industry documents found that the Foundation sponsored a 1965 literature review, set the review's objective, and steered attention toward fat and away from sucrose. (Article) Imagine that you are working for the Sugar Research Foundation in 1965. Some scientists are starting to ask whether eating a lot of sugar increases the risk of heart disease. Your organization is worried that if sugar is blamed, it could hurt sugar sales. Question of interest: Among adults, which dietary factor is more strongly related to heart disease risk: sugar or fat? Desired conclusion: Dietary fat and cholesterol are the main diet-related causes of heart disease. Sugar is not an important cause. Design a study that will likely support your organization's desired conclusion, without fabricating data. Relevant facts: - Heart disease tends to take many years to develop. - Scientists often use blood cholesterol as a risk marker for later heart diseases. - It is challenging to accurately monitor individuals' diets. There are many ways to go about recording fat and sugar intake. Scenario 2: Tobacco Industry The tobacco industry's strategy is very often described not as proving smoking safe, but as manufacturing doubt and uncertainty around the evidence. That general strategy is extensively documented in public health scholarship. (Article) Imagine that you are working for a tobacco company in 1962. More scientists are reporting that cigarette smoking may cause lung cancer. Your company knows it may be hard to prove smoking is harmless, so instead it wants a study that makes the evidence look less clear and less convincing. Question of interest: Among adults, is cigarette smoking clearly associated with respiratory diseases, or could the pattern be explained by other factors? Desired conclusion: The connection between smoking and respiratory diseases is still uncertain, and other explanations may be just as important. Design a study that will likely support your industry's desired conclusion, without fabricating data. Relevant facts: - Lung cancer and other smoking-related diseases usually develop over many years. Shortness of breath, coughing, etc. tend to show up earlier. - Factors like air pollution, occupational exposures (asbestos, arsenic, diesel exhaust, etc.), and radon can affect lung cancer risk.
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Figure 8: Example STATS 8 discussion section activity.

Computation can pose another challenge for teaching authentically: most of my introductory students have never programmed, but I still want them to encounter the tools statisticians use rather than having them use a graphing calculator or work from the same tables “Student” hand-computed for Guinness well over a century ago. At the same time, requiring students to learn R syntax while they are first learning statistical concepts risks making programming, rather than statistical reasoning, the primary difficulty.

My current solution is a Shiny app I created in which students can use a point-and-click menu to insert real R code into an evaluation block they can see and run. Students select the task they want to perform through a graphical interface, but rather than simply returning the result, the application inserts the corresponding R code into a visible evaluation block that students can inspect and run themselves. Figure 10 illustrates this workflow from selecting an analysis to viewing the generated code and its output.

This design preserves important learning experiences for the student while providing significant support for a skill that is not itself a learning objective of the course: remembering the syntax needed to implement that choice in R. It also makes the underlying computation visible rather than hiding it behind a point-and-click menu. My goal is not for the application to substitute for learning R, but to give introductory students an accessible entry point into the computational environment used in statistical practice.

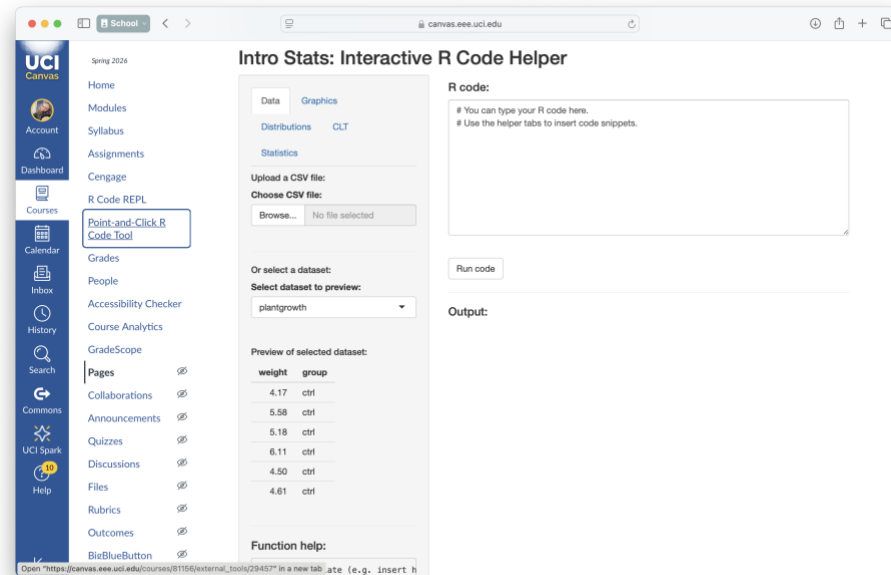


Figure 9: Shiny app embedded into a Canvas page.

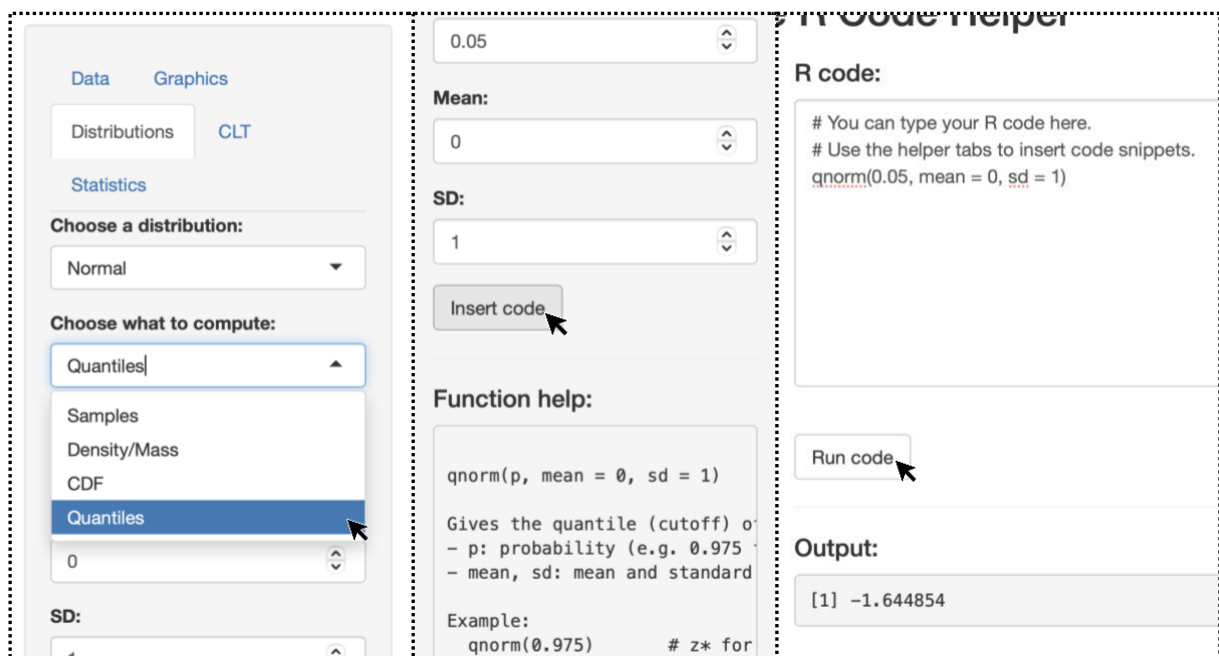


Figure 10: Partial screenshots demonstrating the usage of the Shiny app.

3.4 Inclusive classroom climate

Being uncertain, making an incorrect first attempt, and asking for help are ordinary parts of learning difficult material, but students do not necessarily enter my classroom equally comfortable doing those things. Differences in prior mathematical preparation, previous classroom experiences, access to resources, and confidence can affect whether a student interprets difficulty as a normal part of learning or as evidence that they do not belong in the course. I therefore try to establish early that uncertainty is expected while also identifying barriers that may make it harder for some students to engage with that uncertainty productively.

The welcome survey described in Section 3.2.1 helps me do this before I have even met the class. In addition to asking about students' interests, I ask about their previous mathematical and statistical experience, access to course resources, how they are doing as they begin the term, and what they are worried about in the course. The responses reveal substantial differences in both students' preparation and expectations of themselves. Some students enter having never taken statistics and with little confidence in their mathematical ability, while others have completed AP Statistics or advanced mathematics. They also range from apprehensive about their first exposure to R to excited about learning a programming language. Concerns about grades, exams, workload, and keeping up with the material are also common. Table 2 illustrates some of these different starting points.

Table 2: Students enter STATS 8 with very different starting perspectives.

	Sample student responses to welcome survey
<i>Prior Math Experience & Confidence</i>	"I'm nervous about my own skill level. I really know nothing and I am not confident at all (even with basic algebra). My lack of confidence can get in the way." "I honestly do not have any big concerns about this class, I have taken calculus and upper division mathematics before so I am not too concerned about the difficulty of this class."
<i>Computation</i>	"I honestly am scared to use R." "I am excited to learn a little programming language and a lot of stats and problem solving!"
<i>Expectations for the course</i>	"Like many others, I am concerned about my grade, and I am wondering what exactly I will need to know to do well on the exams. Unfortunately, there is nothing in particular that I am excited about." "I have previously taken AP statistics, and I am excited to go a bit beyond that."

I use these responses to shape the first class meeting rather than simply collecting them for my own information. In STATS 8, I devoted part of the first lecture to the concerns that appeared repeatedly in the survey. I acknowledged the wide variation in students' previous mathematical and statistical preparation and emphasized that the course was designed not to assume prior statistics or programming experience. I also addressed concerns about exams and grades by explaining what I would expect students to be able to do, how those expectations would be communicated, and where students could find support when they encountered difficulty. More generally, I tried to distinguish being unfamiliar with a subject from being incapable of learning it. Difficulty, confusion, and the need to ask questions were experiences I expected students to have in the course, not evidence that they were destined to be bad at statistics.

One student later described the effect of this first-day discussion in an unsolicited email at the end of the term: "Just like all other math classes I've taken, I was already preparing myself to fail with this mindset. But something that surprised me was that in the very first lecture, you addressed all of these feelings. You talked about our feelings and our worries before even getting to the material. You reassured us and set clear expectations for how to succeed in this course."

Relatedly, it is important that students have ways to participate without feeling that every contribution must already be correct. In a lecture of more than 200 students, asking a question publicly can itself be a substantial barrier, particularly when a student is unsure whether their confusion is shared by anyone else. I use Slido (<https://www.slido.com>) during lecture to allow students to submit questions completely anonymously, with no sign-in required. I monitor the questions as I teach and can pause, clarify, or revisit a point when confusion emerges. The anonymity is the key here. I want students to be able to say, "I don't understand this," without

first having to decide whether they are comfortable identifying themselves to the instructor and 200 classmates. The tool also gives me information I would otherwise lose: a student who would never raise their hand can still tell me when an explanation has not worked, or even comment and make jokes with me without having to speak up in a large lecture.

Students have specifically identified this feature as affecting their willingness to participate. One student wrote in an anonymous course evaluation: “I also enjoy how she had an anonymous question site, I feel like that allowed us as students not feel embarrassed when asking a question.”

These practices reflect how I think about inclusion in the classroom. I cannot remove the uncertainty inherent in learning statistics, nor would I want to. I can make uncertainty ordinary, provide students with multiple ways to ask for help while they work through it, and communicate that struggling with a difficult idea is compatible with belonging in the course. Students noticed this; one wrote in an email at the end of the term that “you carry yourself with such compassion and I want to express how this alone helped me overcome my mental block of attempting hard things.”

4 Evidence of Teaching Effectiveness

I use several forms of student feedback to evaluate whether my teaching is accomplishing the goals described in Sections 1 and 3. Table 3 summarizes selected quantitative and qualitative evidence from the two courses for which I have served as instructor of record.

Table 3: Selected student feedback from Instructor of Record courses.

Course / measure	Participation	Notes and Quotes
<u>STATS 8</u> / Official department-activated evaluations	175 of 211 students responded (83%)	Quantitative measures: • <i>Material presented in a way that helped students understand it: 95%</i> agree/strongly agree • <i>Opportunities for learning helped students better understand material: 93%</i> agree/strongly agree • <i>Would recommend instructor: 90%</i> agree/strongly agree • <i>Reported course difficulty level: 15% easy, 0% very easy</i> Quotes: <i>“I think if it was any other professor teaching this course I would drown, but Jessalyn’s personality, methods, and compassionate support is what got me through the rigorous material. It’s encouraging to see the passion that goes into her work and her genuine care for us and our well being does not go unnoticed.”</i> <i>“The most intellectually stimulating parts of the class were learning how statistics can be applied to real world situations and interpreting data to make conclusions. I think after taking this class I actually like stats and now when I see a stat presented to me I do look at it differently.”</i> <i>“It’s super cool that she created an R program into Canvas to use without complications. I also really appreciated it when she had 2 explanations for complex topics: a mathy one and a simplified non-mathy one for people who might not grasp things in that way like me.”</i> <i>“... very clear, straightforward, and easy to understand. Instructor were very clear about her expectation for tests, so that we can prepare well.”</i>

<u>STATS 8</u> / Optional reflective free response question on final exam	204 of 211 students responded (97%)	<i>"I remember Jessalyn at the start of the quarter saying she hopes we see the world differently after this class. I have good news for her, I do."</i> <i>"I believe this class made me a more active participant in data analysis and media literacy, instead of just absorbing whatever information is throughout without question."</i> <i>"As someone who is currently in Chem I found myself constantly thinking about [this class]. It has changed the way I talk about my analyzed data and I believe it has shown in my poster presentations that I'm using intentional language instead of random words I do not understand."</i> <i>"I used to be afraid of any type of statistics... because I was deeply believed I'm just not good at it, at all. After taking this class this quarter, I felt I understood the concepts... now I'm even able to introduce and explain them to my friends."</i>
<u>STATS 67</u> / Official school- activated evaluations	19 of 23 students responded (83%)	Quantitative measures on a scale from 0 (unsatisfactory) to 9 (excellent): • <i>Instructor is accessible and responsive: 8.94/9</i> • <i>Instructor creates an open and fair learning environment: 8.78/9</i> • <i>Instructor encourages students to think: 8.56/9</i> Quotes: <i>"Unlike with other teachers, I wasn't shy on asking her to re-explain concepts or ask her for clarification... very comfortable learning environment."</i> <i>"She is not reading off a script or lesson slides, she is articulating the idea and purpose of each concept in class."</i> <i>"She puts so much effort into preparation for every single subject and even adjusts the material and adds review material if she notices a trend in the class's understanding of a subject."</i>

This feedback reflects several of the teaching practices illustrated earlier in this dossier. In STATS 8, students specifically identified clear expectations for exams, multiple ways of explaining mathematical ideas, anonymous opportunities to ask questions, authentic computation, and the use of real-world contexts as features that supported their learning. These comments are consistent with the broader quantitative results: 95% of respondents agreed or strongly agreed that I presented material in a way that helped them understand it, and 93% agreed or strongly agreed that the opportunities for learning in the course helped them better understand the material.

I am also encouraged that similar themes appear in feedback from STATS 67, despite the substantial differences between the courses. Qualitative comments similarly emphasized comfort asking for clarification, attention to the purpose behind statistical concepts, and my willingness to adjust instruction in response to students' understanding. These results suggest that the principles underlying my teaching have translated across courses with different audiences, scales, and mathematical demands.

The STATS 8 final exam reflections provide a different kind of evidence. I asked students to reflect on whether anything they had learned in the course had affected how they interpreted information or approached problems outside of class. Their responses frequently described changes that closely matched my goals for an introductory statistics course: scrutinizing the methods behind scientific claims rather than accepting conclusions at face value, using more precise language when discussing data in other science courses, questioning statistical claims encountered in the media, and feeling able to explain statistical ideas to others.

One student recalled my telling the class at the beginning of the quarter that I hoped they would see the world differently after taking statistics, and responded, “I have good news for her, I do.” These reflections are particularly meaningful to me because the outcomes students describe extend beyond successful completion of a particular statistical procedure toward the kind of independent statistical reasoning I discuss repeatedly in Section 1.

These positive evaluations did not depend on students perceiving the course as easy. Half of STATS 8 respondents described the course as challenging or very challenging, while only 15% described it as easy and none as very easy. I view that combination, students reporting substantial intellectual challenge while also reporting clarity, support, and increased understanding, as particularly consistent with the balance of rigor and compassion that I aim to create.

5 Efforts to Improve Teaching

I approach my teaching as something to be studied and revised rather than as a fixed set of practices. My development as an instructor thus far has been shaped by two main sources: feedback from students and engagement with scholarship and colleagues in statistics education. I use these sources not only to identify what is working, but also to reconsider teaching choices that may have been based on assumptions rather than evidence.

5.1 Learning from student feedback

I seek feedback about my teaching throughout a course rather than relying only on end-of-term evaluations. Different forms of feedback answer different questions: a welcome survey helps me understand the students entering the course; questions during lecture and patterns in student work help me identify explanations or concepts that need revisiting; targeted post-assessment surveys allow me to evaluate particular teaching choices; and then finally end-of-term evaluations and reflections inform changes to future iterations of a course. For example, I use anonymous Slido questions not only to lower the barrier to participation, as described in Section 3.4, but also as real-time feedback on my instruction: clusters of questions about the same point often prompt me to pause, approach an idea differently, or return to it in a subsequent lecture. Similarly, after introducing the proficiency-based exam preparation described in Section 3.1, I asked students directly about the clarity of the exam expectations rather than assuming that the materials were functioning as I intended.

Feedback sometimes prompts larger changes in my teaching. In my first offering of STATS 67, I replaced traditional in-class exams with timed take-home quizzes because I had come to view exams primarily as unnecessary stressors. A student commented in the end-of-term evaluations that “assessments don’t matter as much as they are open book... I personally feel there is some merit to a ‘traditional’ exam.” That comment made me reconsider an assumption I had made about compassionate teaching. I began engaging with statistics education literature around that time through a reading group with Professor Mine Dogucu, who was then a Professor of Teaching at UCI. Reading about specifications grading (Nilson, 2015) clarified something I had not yet articulated: my problem with most assessment is not its *existence*, but its *opacity*. There is an important difference between an exam that is difficult because the material is challenging and one that is difficult because students do not know what to expect. Separately, reading about how exams structure learning helped me reconceive assessments not as adversarial tools of post-hoc evaluation, but as tools that direct time and effort toward productive learning (Gibbs & Simpson, 2004).

I took these lessons with me into my next course, STATS 8. I returned to closed-book exams, but I paired them with detailed lists of expected competencies that made explicit what students should be able to do as described in Section 3.1. I have continued refining this approach in subsequent courses, including my 2026 offering of STATS 67. This progression, from removing exams, to reconsidering their purpose, to redesigning them around

transparent expectations, is one example of how student feedback and teaching scholarship interact in my development as an instructor.

5.2 Engagement with teaching scholarship

My engagement with education scholarship began through a reading group led by Professor Mine Dogucu, then a Professor of Teaching at UCI. Since then, I have continued to follow work in statistics education, particularly in the *Journal of Statistics and Data Science Education*, *Statistics Education Research Journal*, *Teaching Statistics*, and *Technology Innovations in Statistics Education*. One particularly important influence on my teaching has been the American Statistical Association's *Guidelines for Assessment and Instruction in Statistics Education* (GAISE) College Report (Carver, et al., 2016). Its emphasis on teaching statistical thinking, fostering active learning, using real data and meaningful contexts, emphasizing conceptual understanding rather than procedural calculation, and using technology to explore concepts closely informs the kind of introductory statistics course I try to create. Work by Garfield, Ben-Zvi, and colleagues on statistical reasoning and the Statistical Reasoning Learning Environment has similarly influenced how I think about moving students beyond performing procedures toward making and justifying statistical decisions (Garfield & Ben-Zvi, 2009; Garfield, et al., 2008). Gelman and Nolan's *Teaching Statistics: A Bag of Tricks* (2017), particularly its use of demonstrations, real data, statistical literacy, and open-ended activities, provides another model for making statistical ideas concrete without reducing them to recipes.

Scholarship on assessment has been especially important as my own approach to assessment has evolved, as discussed in Section 5.1. Work on assessment in statistics education, including Gal and Garfield's edited volume *The Assessment Challenge in Statistics Education* (1997), has reinforced my interest in assessments that elicit conceptual understanding and statistical reasoning rather than recognition of a familiar procedural template.

Although my engagement with discipline-based education research has primarily been through statistics education, many of the questions that arise in my teaching overlap with scholarship in undergraduate mathematics education. I am particularly interested in work on the relationship between procedural and conceptual knowledge and on helping students develop mathematical reasoning rather than treating mathematics as a collection of procedures (Hiebert & Lefevre, 1986; Star, 2005). These questions arise directly in my teaching of statistics. For example, in STATS 67, it is not only important to be able to carry out a derivation, but also to understand what the resulting mathematical properties tell them about a statistical method and to use those properties to justify a decision. I am interested in engaging more deeply with mathematics education scholarship as I continue teaching courses across the mathematics and statistics curriculum.

Although my research to date has primarily been methodological statistics rather than statistics education, I increasingly see my teaching as another area in which I want to work scholarly. My teaching has raised questions for me about, for example, whether explicit proficiency statements change how students prepare for assessments; how computational scaffolds can expose students to authentic statistical tools without making programming syntax an unnecessary initial barrier; and whether the statistical literacy students develop in an introductory course transfers to how they evaluate evidence in other courses, scientific literature, and everyday life. In the future, I am interested in developing some of these questions into statistics education research of my own and in collaborating with colleagues working in statistics and mathematics education.

6 Teaching Interests

My teaching experience to date has been primarily in statistics, but I would also be enthusiastic about contributing to lower-division mathematics courses. My training in statistics, together with my undergraduate minor in physics, has given me a strong mathematical foundation in calculus, linear algebra, probability, and related areas, and I regularly draw on this foundation in both my research and teaching. For example, STATS

67 requires students to use single- and multivariable calculus and linear algebra to derive properties of probability distributions and statistical estimators, while my graduate teaching and research make extensive use of linear algebra, calculus, and differential equations. I would therefore be comfortable teaching courses such as introductory calculus, multivariable calculus, linear algebra, and introductory differential equations, in addition to probability and statistics courses across the curriculum. At the upper-division level, my strongest teaching interests are in statistics, including probability, statistical modeling, mathematical statistics, Bayesian statistics, and statistical computing; my graduate teaching experience in statistical computing and generalized linear models has given me experience supporting students with material at this level.

I would approach all of these courses with many of the same principles that guide my current teaching. My background in statistics and physics has made me particularly interested in the connections between mathematical ideas and the scientific problems they describe. In calculus and differential equations, I would want students not only to carry out computations but also to connect those computations to the behavior of the quantities they describe and to applications in the sciences. In linear algebra, I would similarly emphasize the relationship between formal operations and their geometric, computational, and statistical interpretations. Very generally, I am interested in helping students develop the ability to move between mathematical procedures, conceptual understanding, and applications.

Beyond contributing to teaching needs, I would be interested in participating in broader educational work around curriculum development and evidence-based pedagogy. Developing resources such as *deitaset*s and my introductory statistics computing tool has made me particularly interested in creating and evaluating teaching materials that can be used beyond a single course. As I gain experience across a broader range of mathematics and statistics courses, I would welcome opportunities to contribute to curriculum development and other collaborative educational initiatives.

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